

TASK 1

Introduction to IoT & Device Communication

Decodelabs IoT Internship

Goal: Understand the basic concept of IoT and how devices communicate with each other.

1. Introduction

The Internet of Things (IoT) is a network of physical devices embedded with sensors, software, and connectivity that enables them to collect and exchange data over the internet. These devices range from everyday household appliances to large industrial machines, all communicating without direct human intervention.

IoT is reshaping every industry by enabling smarter decisions through real-time data. According to Statista, over 15 billion IoT devices were connected globally in 2023, projected to exceed 29 billion by 2030. Understanding how these devices are structured and how they communicate is foundational to any IoT project.

2. IoT Architecture

IoT systems follow a three-layer architecture: the Device Layer, the Network Layer, and the Cloud Layer. Each layer plays a distinct role in the end-to-end flow of data.

Layer	Components	Function
Device Layer	Sensors, actuators, microcontrollers (Arduino, Raspberry Pi)	Collects physical data from the environment
Network Layer	Wi-Fi, Bluetooth, Zigbee, Z-Wave, MQTT, 4G/5G	Transmits data between devices and cloud
Cloud Layer	AWS IoT, Google Cloud IoT, Azure IoT Hub	Stores, processes, analyses data; triggers automated actions

2.1 Device Layer

Sensors measure real-world parameters such as temperature, humidity, and motion. Actuators perform physical actions like opening a valve or switching a light. Microcontrollers (Arduino, Raspberry Pi) process sensor inputs and initiate communication.

2.2 Network Layer

Different protocols are chosen based on range, power, and bandwidth requirements:

- **Wi-Fi** – High bandwidth; ideal for home devices and video streaming.
- **Bluetooth / BLE** – Short-range; used in wearables and smart accessories.
- **Zigbee / Z-Wave** – Low power mesh networking; common in smart home sensors.
- **MQTT** – Lightweight publish-subscribe protocol; widely used for IoT data.
- **4G / 5G** – Long-range cellular; used in industrial IoT and vehicle tracking.

2.3 Cloud Layer

Cloud platforms (AWS IoT Core, Google Cloud IoT, Azure IoT Hub) receive, store, and process device data. They host dashboards and APIs that allow users to monitor and control devices remotely, and can trigger automated responses based on defined rules.

3. Real-World IoT Use Cases

■ Smart Homes

Connected thermostats (Nest), smart lighting (Philips Hue), video doorbells (Ring), and voice assistants (Amazon Alexa) automate home management and improve energy efficiency.

■ Smart Healthcare

Wearables monitor heart rate, blood oxygen, and glucose continuously. IoT hospital beds and infusion pumps improve patient safety and reduce manual nurse checks.

■ Smart Agriculture

Soil moisture sensors, drone-based crop monitoring, and automated irrigation let farmers optimise water usage and maximise crop yields.

■ Industrial IoT (IIoT)

Predictive maintenance sensors on factory machines detect anomalies before breakdowns, reducing downtime and maintenance costs significantly.

■ Smart Transportation

GPS trackers, traffic sensors, and connected vehicles enable real-time fleet management, route optimisation, and autonomous driving support.

■ Smart Cities

IoT-powered streetlights, waste bins, and water meters cut resource consumption and improve urban services for millions of residents.

4. Data Flow in an IoT System

Data in an IoT system travels through five clearly defined stages from physical sensing to a final automated action:

Stage	Description
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1. Sensing	Sensor detects a physical parameter (e.g. temperature = 32°C).
2. Data Capture	Microcontroller reads the sensor value and converts it to a digital signal.
3. Transmission	Data is sent over a protocol (e.g. MQTT over Wi-Fi) to a gateway or cloud.
4. Processing	Cloud platform stores the data and applies logic (e.g. if temp > 30°C, alert).
5. Action	Actuator responds (e.g. AC turns on) or a notification is sent to the user.

This pipeline — from physical sensing to cloud-based intelligence — is what makes IoT powerful. Each stage must be optimised for latency, reliability, and security.

5. Key Skills Developed

- **IoT Fundamentals** – Understanding what IoT is, its history, and its ecosystem.
- **System Architecture** – Mapping devices, networks, and cloud components for a use case.
- **Protocol Knowledge** – Knowing Wi-Fi, MQTT, Zigbee, BLE and when to apply each.
- **Data Flow Thinking** – Tracing data from sensor to action and identifying bottlenecks.
- **Use-Case Analysis** – Evaluating real-world applications to inform IoT design decisions.

6. Conclusion

IoT connects physical devices to the internet, enabling them to share data and automate actions that improve everyday life. The three-layer architecture (Device → Network → Cloud) provides a clear model for understanding any IoT system. Real-world applications — smart homes, healthcare, agriculture, industry — confirm that IoT is a global infrastructure, not just a trend.

The fundamentals covered in this task — architecture, protocols, data flow, and use cases — form the essential foundation for all further IoT development tasks in this internship.